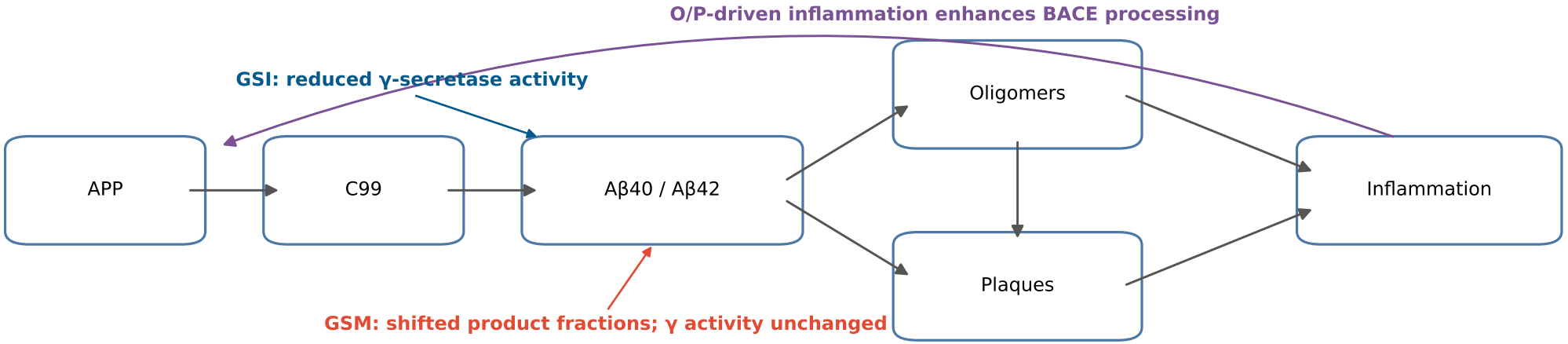
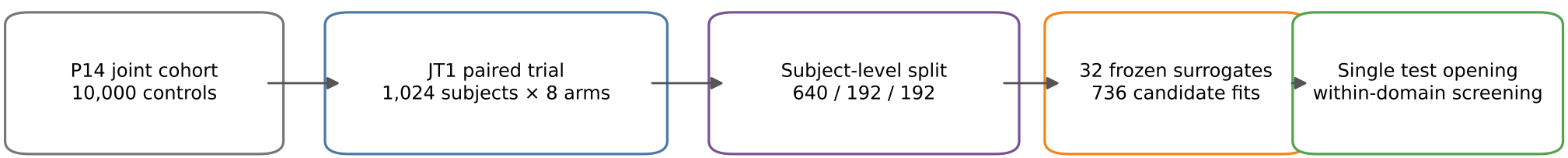
Mechanistic-to-surrogate pathway-level digital twin

A





Numerical audit → paired mechanistic responses → frozen domain guard